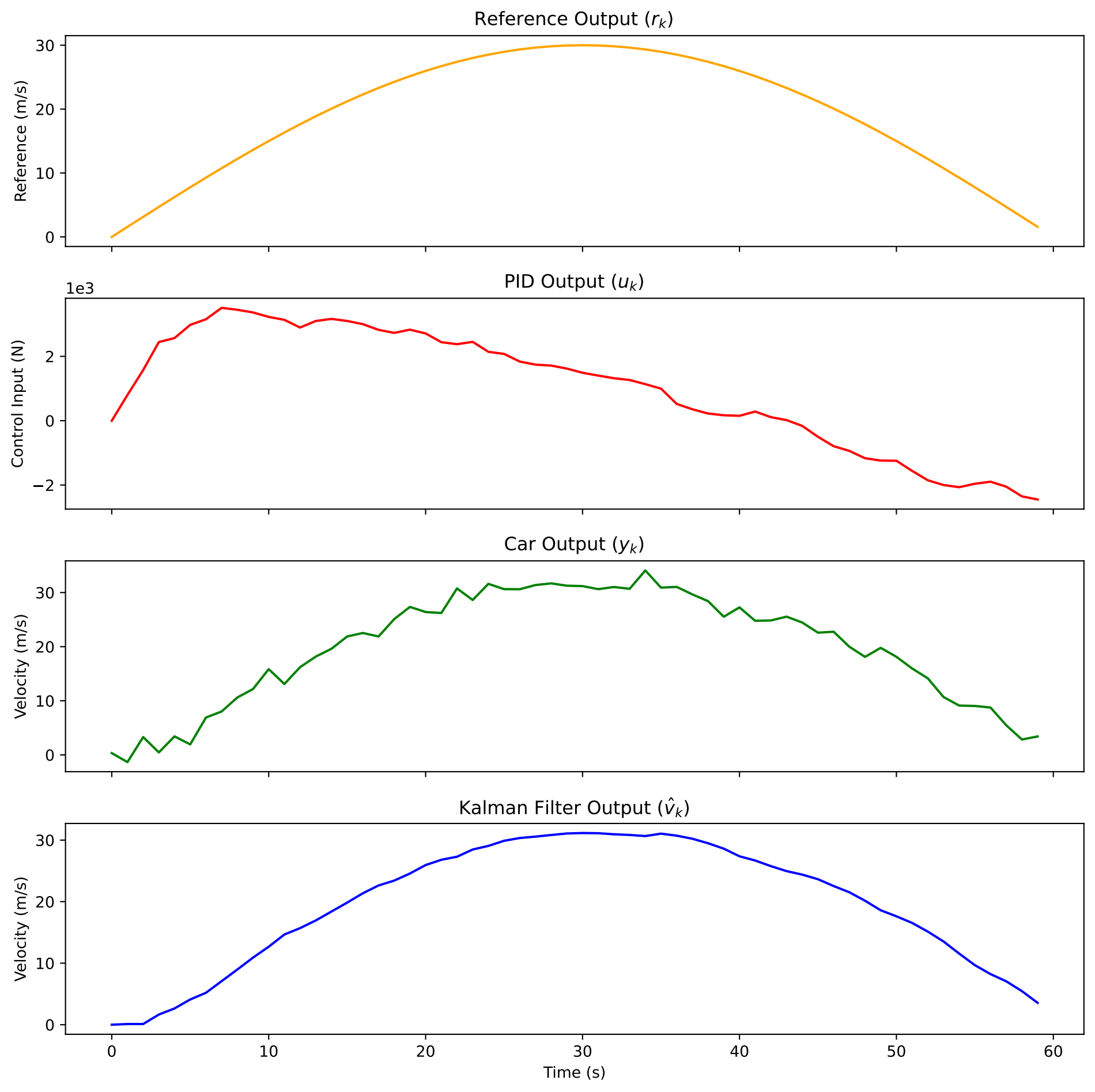
Reference Output (r_k)
Reference (m/s)
30
20
10
0
PID Output (u_k)
1e3
Control Input (N)
2
0
−2
Car Output (y_k)
Velocity (m/s)
30
20
10
0
Kalman Filter Output ($\hat{v}_k$)
Velocity (m/s)
30
20
10
0
0
10
20
30
40
50
60
Time (s)